

Figure 2

Romana Limberger, Uriah Daugaard et al.

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1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(patchwork)
library(here)
library(lme4)
library(lmerTest)
```

1.2 Data

```
load(here("Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"))

complete.forecasts.taxa <- complete.forecasts %>%
  dplyr::filter(
    forecast.method == "Simplex EDM",
    Target != "Total_biomass",
    Target != "Mean_oxygen"
  )
```

1.3 Functions

```
conf_interval <- function(model, nd, response, renameResponse, cmult=1.96){
  nd[,response] <- predict(model, nd, re.form=NA)
  mm <- model.matrix(terms(model),nd)
  pvar1 <- diag(mm %*% tcrossprod(vcov(model),mm))
  nd <- nd %>%
    dplyr::mutate(lower=!!as.name(response)-cmult*sqrt(pvar1),
                  upper=!!as.name(response)+cmult*sqrt(pvar1),
                  Response=renameResponse) %>%
    dplyr::rename(fit=!!as.name(response))
  return(nd)
}
```

2 Figure 2

2.1 Models

```
## acf vs exp
m.acf.exp <-
  lmer(
    acf_lag5 ~ median_richness_center + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## cv vs exp
m.cv.exp <-
  lmer(
    cv.res.log10 ~ median_richness_center + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

### pe vs rmse
m.pe.exp <-
  lmer(
```

```

    pe5 ~ median_richness_center + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## acf vs RMSE
m.acf.rmse <-
  lmer(
    RMSE ~ acf_lag5 + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

## CV vs RMSE
m.cv.rmse <-
  lmer(
    RMSE ~ cv.res.log10 + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

### pe vs rmse
m.pe.rmse <-
  lmer(
    RMSE ~ pe5 + light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator),
    complete.forecasts.taxa
  )

```

2.2 Figure

```

newdat.m.exp <-
  expand.grid(
    median_richness_center = seq(
      min(complete.forecasts.taxa$median_richness_center),
      max(complete.forecasts.taxa$median_richness_center),

```

```

    length.out = 100
  ),
  light_treatment = unique(complete.forecasts.taxa$light_treatment)
)

newdat.m.acf <-
  expand.grid(
    acf_lag5 = seq(
      min(complete.forecasts.taxa$acf_lag5),
      max(complete.forecasts.taxa$acf_lag5),
      length.out = 100
    ),
    light_treatment = unique(complete.forecasts.taxa$light_treatment)
  )

newdat.m.cv <-
  expand.grid(
    cv.res.log10 = seq(min(log10(
      complete.forecasts.taxa$cv.res
    )),
    max(log10(
      complete.forecasts.taxa$cv.res
    )),
    length.out = 100),
    light_treatment = unique(complete.forecasts.taxa$light_treatment)
  )

newdat.m.pe.rmse <-
  expand.grid(
    pe5 = seq(
      min(complete.forecasts.taxa$pe5),
      max(complete.forecasts.taxa$pe5),
      length.out = 100
    ),
    light_treatment = unique(complete.forecasts.taxa$light_treatment)
  )

```

```

## acf vs exp
newdat.acf.exp <-
  conf_interval(m.acf.exp, newdat.m.exp, "acf_lag5", "Autocorrelation at lag 5")

## cv vs exp
newdat.cv.exp <-
  conf_interval(m.cv.exp, newdat.m.exp, "cv.res.log10", "cv.res.log10")

## acf vs RMSE
newdat.m.acf.rmse <-
  conf_interval(m.acf.rmse, newdat.m.acf, "RMSE", "Forecast error (RMSE)")

## CV vs RMSE
newdat.m.cv.rmse <-
  conf_interval(m.cv.rmse, newdat.m.cv, "RMSE", "Forecast error (RMSE)")

### pe vs exp design
newdat.m.exp <-
  conf_interval(m.pe.exp, newdat.m.exp, "pe5", "Permutation entropy")

### pe vs rmse
newdat.m.pe.rmse <-
  conf_interval(m.pe.rmse, newdat.m.pe.rmse, "RMSE", "Forecast error (RMSE)")

fig2exp.acf <- complete.forecasts.taxa %>%
  ggplot(aes(median_richness_center, acf_lag5, col = light_treatment)) +
  geom_ribbon(
    data = newdat.acf.exp,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    )
  )

```

```

    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.acf.exp,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Centered median realized species richness",
       y = "Autocorrelation at lag 5",
       col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "none")

fig2exp.cv <- complete.forecasts.taxa %>%
  ggplot(aes(median_richness_center, cv.res.log10, col = light_treatment)) +
  geom_ribbon(
    data = newdat.cv.exp,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.cv.exp,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Centered median realized species richness",
       y = "log10(Coefficient of variation CV)",
       col = "Light conditions") +

```

```

theme_bw() +
theme(legend.position = "bottom")

fig2acf.rmse <- complete.forecasts.taxa %>%
  ggplot(aes(acf_lag5, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.acf.rmse,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.acf.rmse,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Autocorrelation at lag 5", y = "Forecast error (RMSE)",
       col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "none")

fig2cv.rmse <- complete.forecasts.taxa %>%
  ggplot(aes(cv.res.log10, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.cv.rmse,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
  ),

```

```

    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.cv.rmse,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "log10(Coefficient of variation CV)",
       y = "Forecast error (RMSE)",
       col = "Light conditions",
       tag = "e") +
  theme_bw() +
  theme(legend.position = "none")

fig2exp.pe <- complete.forecasts.taxa %>%
  dplyr::filter(Target != "Total_biomass",
                Target != "Mean_oxygen",
                forecast.method == "Simplex EDM") %>%
  ggplot(aes(median_richness_center, pe5, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.exp,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.exp,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +

```

```

labs(x = "Centered median realized species richness",
     y = "Permutation entropy",
     col = "Light conditions") +
theme_bw() +
theme(legend.position = "none")

fig2pe.rmse <- complete.forecasts.taxa %>%
  dplyr::filter(Target != "Total_biomass",
                Target != "Mean_oxygen",
                forecast.method == "Simplex EDM") %>%
  ggplot(aes(pe5, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.pe.rmse,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.pe.rmse,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(y = "Forecast error (RMSE)", x = "Permutation entropy",
       col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "none")

fig2acf.rmse + fig2cv.rmse + fig2pe.rmse +
  fig2exp.acf + fig2exp.cv + fig2exp.pe +
  plot_layout(ncol = 3) +
  plot_annotation(tag_levels = "a") &

```

```
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill"))
```

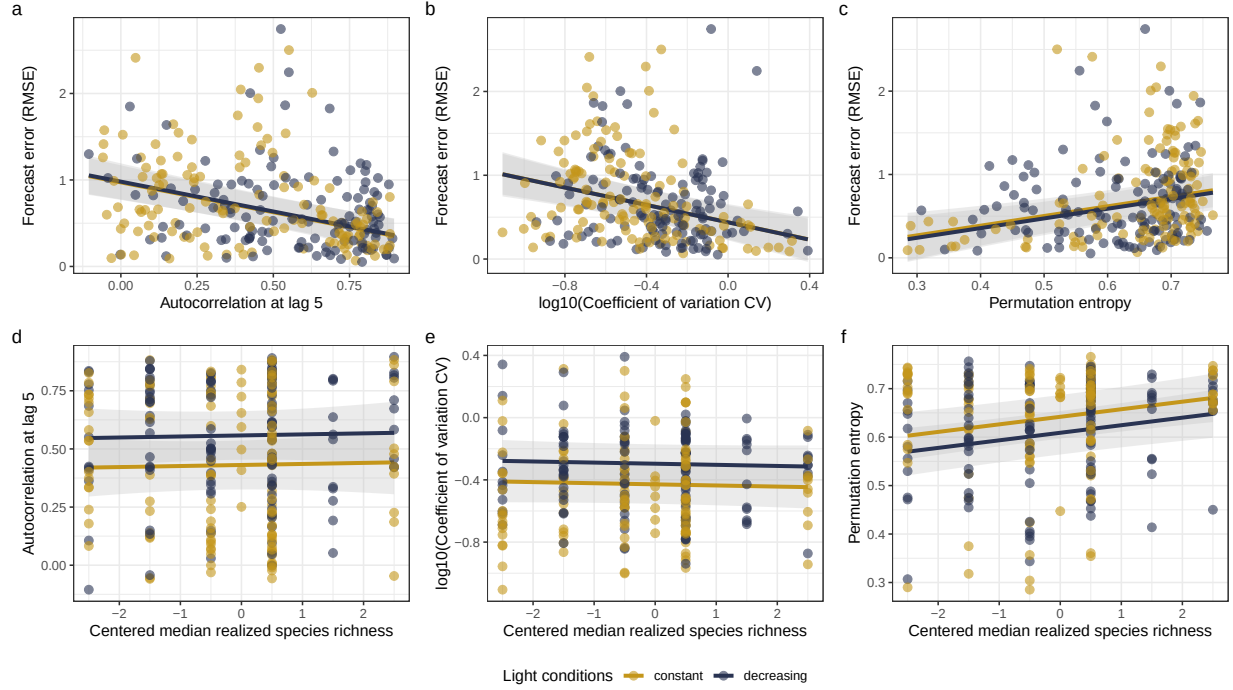


Figure 2: Relationship of time series metrics with forecast error (a-c), and with experimental treatments (d-f). Time series metrics are (a, d) autocorrelation at lag 5, (b, e) coefficient of variation as a measure of temporal variability, and (c, f) permutation entropy as a measure of time series complexity. All variables are unitless. Time series metrics were calculated for 253 taxa time series (across all bottles). Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Taxon, bottle, and incubator were included as random effects in the model. Forecast error is based on forecasts computed with Simplex EDM. Note that in (b) and (e) the coefficient of variation was log10-transformed so that linear model assumptions were met.

3 Versions of R and of the R-packages used

R version: R version 4.4.1 (2024-06-14 ucrt).

Base packages: stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: lmerTest 3.1-3, lme4 1.1-36, Matrix 1.7-0, here 1.0.1, patchwork 1.3.0, lubridate 1.9.4, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.4, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.5.1, tidyverse 2.0.0, knitr 1.49.

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